

Chapter 5

Statistical Data Treatment and Evaluation

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 5 Problems:

5-4, 5-6, 5-31

Statistical analysis only reveals information that is already present in a data set (in other words no new information). It allows us to characterize data and make decisions about data quality and interpretation.

The Confidence Interval for the mean is the range of values within which the population mean μ is expected to lie with a certain probability.

Sometimes expressed as a **Confidence Limit** for example 95% probable that the true population mean falls within an interval of $8.25 \pm .15$, therefore the probability that the population mean lies within the interval 8.10-8.40 is 95%

Finding the Confidence Interval when σ is Unknown

$$CL_{95\%} = \frac{t * s}{\sqrt{N}}$$

Example 5-3

The following results were obtained in the replicate determinations of blood alcohol content: 0.084, 0.089, 0.079. Calculate the 95% CL

Statistical Aids to Hypothesis Testing

A null hypothesis postulates that two or more observed quantities are the same. It assumes the two values be compared are the “same” in the absence of systematic error.

Examples of Hypothesis tests include the comparison of

- 1) The mean of an experimental data set with what is believed to be the true value (t-test)
 - 2) Two experimental mean values to each other (paired t-test)
 - 3) The standard deviation from two or more sets of data (F-test)
 - 4) Tests an outlier to determine if it can be rejected from a set of data (Q-test)
1. Comparing an Experimental Mean with a Known Value

Use a hypothesis test to draw conclusions about the population mean μ and its nearness to the know value μ_o (accepted value)

$$H_0: \mu = \mu_o \text{ (accepted value)}$$

$$H_a: \mu \neq \mu_o$$

If $X_{\text{exp}} > X_{\text{crit}}$ then the null hypothesis fails, so there is a significant difference

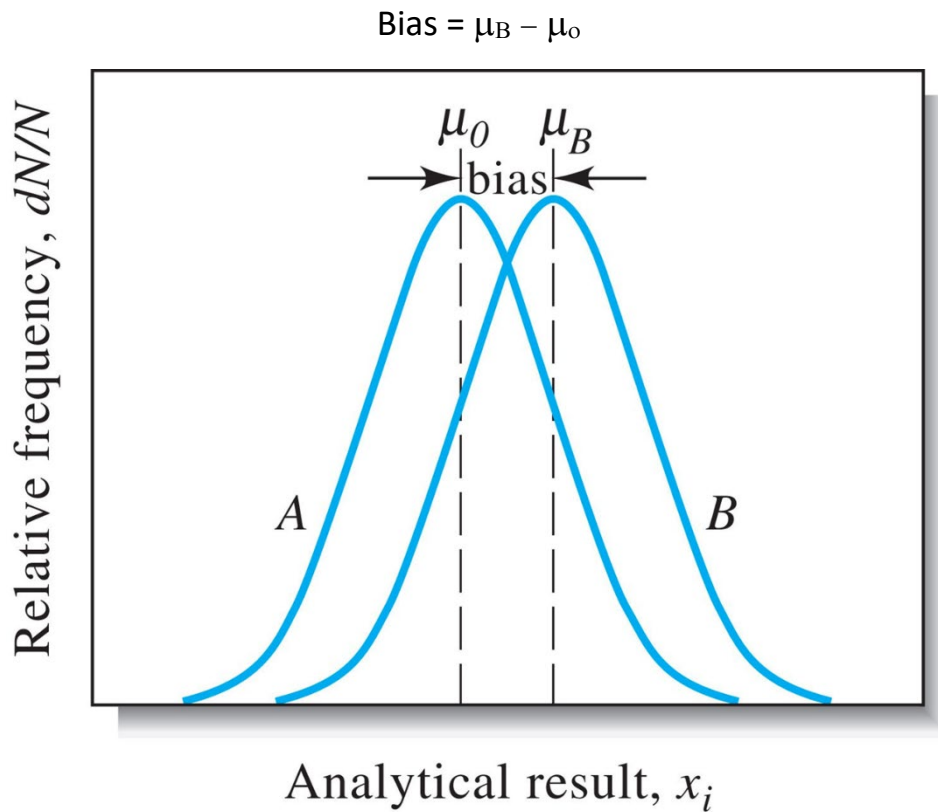
Small Sample t Test

1. State the null hypothesis $H_0 : \mu = \mu_o$
2. Form the test statistic

$$t_{exp} = \frac{|\bar{x} - \mu_o|}{s} * \sqrt{N}$$

3. State the alternative hypothesis H_a and determine the rejection region
For $H_a: \mu \neq \mu_o$, reject H_0 if $t \geq t_{crit}$ (one-tailed test)
(goto table 5-3)

Consider testing for systematic error in an analytical method **Figure 5-3**



Example 5-5

A new procedure for the rapid determination of sulfur in kerosenes was tested on a sample known from its method of preparation to contain 0.123% S ($\mu_o = 0.123\% \text{ S}$). The results for % S were 0.112, 0.118, 0.115, and 0.119. Do the data indicate that there is a bias in the method at the 95% CL?

$$H_o: \mu = 0.123\% \text{ S}$$

$$H_a: \mu \neq 0.123\% \text{ S}$$

$$\text{Mean} = 0.116\% \text{ S}$$

$$s = 0.003\% \text{ S}$$

Pooling Data to Improve the Reliability of s

$$t_{exp} = \frac{|\bar{x}_A - \bar{x}_B|}{s_{pooled}} * \sqrt{\frac{N_A * N_B}{N_A + N_B}}$$

2. Comparison of Two Experimental Means

The t Test for Differences in Means

$$s_{pooled} = \sqrt{\frac{[(s_A)^2 * (N_A - 1)] + [(s_B)^2 * (N_B - 1)]}{N_A + N_B - 2}}$$

Where $N_A + N_B - 2$ = Degrees of freedom for the pooled dataset

Example 5-6

3. A glass containing red wine and an open bottle were analyzed for their alcohol content in order to determine whether the wine in the glass came from the bottle. On the basis of six analyses, the average content of the wine from the glass was 12.61% ethanol. For analysis of wine from the bottle gave a mean of 12.53%. The 10 analyses yielded $s_{\text{pooled}} = 0.07\%$. Do the data indicate a difference between the wines?

3. Comparison of Variances

The F test is based on the null hypothesis that the two population variances under consideration are equal

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$F_{exp} = \frac{(s_{larger})^2}{(s_{smaller})^2}$$

F is calculated and compared with the critical value of F at the desired significance level.

Example 5-8

4. A standard method for the determination of carbon monoxide level in gaseous mixtures is known from many hundreds of measurements to have a standard deviation of 0.21 ppm. A modification of the method yields a value for $\bar{x}$ of 0.15 ppm CO for a pooled data set with 12 degrees of freedom. A second modification also based on 12 degrees of freedom, has a standard deviation of 0.12 ppm CO. Is either modification significantly more precise than the original?

We test the null hypothesis $H_0 : \sigma_{\text{std}}^2 = \sigma_1^2$

	Accepted Method	Method 1	Method 2
Std Dev.	0.21	0.15	0.12
N	∞	13	13
DF	∞	12	12

For the first method

For the second method

4. Detection of Gross Errors

The Q test

Tests an outlier to determine if it can be rejected from a set of data.

$$Q_{exp} = \frac{|x_{?} - x_{nearest}|}{w}$$

If Q is greater than Q_{crit} then the questionable result can be rejected with the indicated degree of confidence

Example 5-11

The analysis of a city drinking water for arsenic yielded values of 5.60, 5.64, 5.70, 5.69, and 5.81 ppm. The last value appears anomalous: should it be rejected at the 90 % confidence level?